

VLSI Design Internship – Task 3

Verilog RTL Design of Sequential Circuits and Flip-Flops

Detailed Project Documentation Report

1. Introduction

Sequential circuits are digital circuits whose outputs depend on both present inputs and previous states. Unlike combinational circuits, sequential circuits use memory elements and clock signals to store and process data. This project focuses on implementing D Flip-Flops, JK Flip-Flops, Registers, and Counters using Verilog HDL.

2. Objectives

- Understand sequential logic concepts.
- Learn flip-flop operation and timing behavior.
- Design RTL code using Verilog HDL.
- Develop testbenches for verification.
- Perform simulation and waveform analysis.

3. D Flip-Flop

A D Flip-Flop stores one bit of data. On every positive edge of the clock signal, the output Q follows input D. It is widely used in registers, memories, and state machines.

4. JK Flip-Flop

A JK Flip-Flop is an improved version of the SR Flip-Flop. Depending on the values of J and K, the output can hold, reset, set, or toggle.

5. Registers

Registers are collections of flip-flops used to store multiple bits of data. A 4-bit register stores four bits simultaneously and updates on clock edges.

6. Counters

Counters are sequential circuits that change state according to clock pulses. A 4-bit up counter counts from 0000 to 1111 and then repeats.

7. Tools Used

- Verilog HDL
- EDA Playground
- Icarus Verilog
- GTKWave
- GitHub

8. Testbench Development

Testbenches are used to verify RTL functionality. Clock signals, input stimuli, and expected outputs are applied to validate the design.

9. Simulation and Waveform Analysis

Simulation confirms correct circuit behavior. GTKWave is used to observe clock transitions, input changes, and output responses. Waveforms help identify timing and logic issues.

10. Expected Deliverables

1. D Flip-Flop Verilog Code and Testbench
2. JK Flip-Flop Verilog Code and Testbench
3. 4-Bit Register Verilog Code and Testbench
4. 4-Bit Up Counter Verilog Code and Testbench
5. Waveform Screenshots
6. README Documentation
7. GitHub Repository Link

11. Conclusion

Task 3 provides hands-on experience in RTL design, verification, and simulation of sequential circuits. The concepts learned in this project form the foundation for advanced VLSI design, FPGA development, digital system design, and hardware verification.